MIL-V-50081F
14 October 1976
SUPERSEDING
MIL-C-0050081E(EA)
6 August 1976
MIL-C-50081B
30 September 1964

MILITARY SPECIFICATION

VALVE, INLET, CHEMICAL-BIOLOGICAL MASK, C20R1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of inlet valve for chemical-biological masks.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Edgewood Arsenal, Attn: SAREA-TS-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

C5-2-1014	-	Valve, Inlet, Chemical-Biological Mask, C20R1.
B136-14-170-3	-	Air Leakage Tester, Flow Type, Q211, Assembly.
B136-44-1-2	-	Resistance Indicator, Inlet and Outlet Valve Q212, Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to specifications and drawings forming a part of this specification.

3.2 Manufacture and assembly. The inlet valve shall be manufactured and assembled as specified on Drawing C5-2-1014, and subsidiary drawings listed thereon.

3.3 Valve leakage. Leakage through the valve shall not exceed 150 milliliters per minute at an air pressure differential of one inch of water when tested as specified in 4.4.4.1.

3.4 Airflow resistance. Resistance through the valve shall not exceed 12 millimeters of water when tested at a constant airflow rate of 42.5 liters per minute as specified in 4.4.4.2.

3.5 Preproduction sample. Prior to the start of regular production, a preproduction sample of inlet valves shall be produced in accordance with this specification for examination and tests (see 4.3). Any change in materials, operating conditions or dies shall require a new preproduction sample.

3.6 Workmanship. The inlet valve shall be free from foreign matter and damage such as tears in the disk and dents which will impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of ten inlet valves shall be produced using the same methods, materials and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The sample valves shall be inspected for all requirements of the drawings and this specification.

4.3.2.2 For tests. Preproduction sample valves shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction valves shall meet the examination and tests as specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A regular production lot shall consist of the valves produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions.

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4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample valves from regular production lots shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Valve, inlet, chemical-biological mask, C20R1 (Dwg C5-2-1014).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.10 percent defective	
101	Valve leakage	4.4.4.1
102	Airflow resistance	4.4.4.2
	AQL 1.0 percent defective	
103	Components missing, incorrect, or incorrectly assembled	
#104	Diameter (1.519) incorrect	
#*105	Length (0.210) incorrect	
#106	Diameter (1.537) incorrect	
107	Three beads incorrect	
108	Marking missing, incorrect, or incorrectly located	
109	Workmanship (see 3.6)	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing, inadequate, or incorrect	

#Reference Drawing C5-2-1016.
*Tool-controlled (may be inspected periodically).

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(b) Ring, retainer (Dwg C5-2-981).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
*101	Diameter (1.750) incorrect
*102	Diameter (1.573) incorrect
103	Height (0.275) incorrect
104	Burrs

*Tool-controlled (may be inspected periodically).

4.4.4 Tests.

4.4.4.1 Valve leakage. Leakage of the inlet valve shall be determined with the Q211 Air Leakage Tester (Dwg B136-14-170-3).

4.4.4.2 Airflow resistance. Resistance of the inlet valve shall be determined with the Q212 Resistance Indicator (Dwg B136-44-1-2).

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking for interplant shipment (see 6.4). Inlet valves shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in accordance with FED-STD-123.

5.2 Repair parts. When inlet valves are procured for storage and issue as repair parts, preservation, packaging, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. This specification describes an inlet valve intended to be used as a component of the ABC-M17 and M17A1 Chemical-Biological Masks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

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(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Part 2, QSTAGS List). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.4 Interplant shipment. Interplant shipment is shipment of supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or prime contractor, from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodian:

Army - EA
Navy - SH
Air Force - 99

Review activities:

Army - MD
Navy - SH

User activities:

Navy - MC

Preparing activity:

Army - EA

Project No. 4240-0340

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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